Jonah Kudler-Flam

Curriculum Vitae

Contact

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Employment

Institute for Advanced Study, Princeton, NJ — 2022 - 2027
Marvin L. Goldberger Member

Princeton University, Princeton, NJ — 2022 - 2025
Affiliated PCTS Postdoctoral Fellow

Education

University of Chicago — 2017 - 2022
M.S. in Physics
Ph.D. in Physics
Advisor: Shinsei Ryu

Princeton University — 2021 - 2022
Visiting Student Research Collaborator

Massachusetts Institute of Technology — 2020 - 2021
Visiting Student
Host: Hong Liu

Colgate University — 2013 - 2017
A.B. in Astronomy/Physics *magna cum laude*

Undergraduate Research Mentorship

Research Advisor to Undergraduate Students

Paris Zhang, University of California, Santa Barbara — 2026–Present
Third/fourth-year undergraduate. Project on von Neumann algebras and observers in de Sitter toy models.

Badis Labbedi, University of Chicago — 2026–Present
Second/third-year undergraduate. Project on the gravitational path integral and ensemble averaging.

Bowen Ouyang, Shanghai Jiao Tong University — 2025–2026
Fourth-year undergraduate. Co-author on an operational meaning for the logarithmic negativity, arXiv:2607.01320 (under review).
Next stop: Physics Ph.D., Brandeis University.

Hyaline Chen, Princeton University — 2024
Project on free probability in dual-unitary quantum circuits, published in *Physical Review B*. Awarded the PACM Undergraduate Certificate Prize for Best Research Project.
Next stop: Physics Ph.D., Harvard University.

Chang-Han Chen, Massachusetts Institute of Technology — 2020–2021
Senior thesis on quantum information in thermalizing systems.

Next stop: Physics Ph.D., University of California, Berkeley.

Founder and Organizer, Theory Group Meeting 2019–2022
University of Chicago
Created a weekly, low-pressure venue for undergraduate and graduate students to present work in progress and practice research talks with supportive peer feedback.

Teaching Experience

Prison Teaching Initiative Instructor 2023–Present
Princeton University, Princeton, NJ
PHYS-130: *Astronomy* (Fall 2025, East Jersey State Prison; Spring 2025, Northern State Prison; Fall 2024, East Jersey State Prison)
MATH-110: *Statistics* (Spring 2024, South Woods State Prison)
MATH-015: *Basic Mathematics* (Fall 2023, South Woods State Prison)

Prison Teaching Initiative Course Coordinator 2024–2025
Princeton University, Princeton, NJ
Coordinated **PHYS-130**: *Astronomy* (Fall 2024, Spring 2025). Redesigned the curriculum and mentored first-time instructors.

Guest Lecturer 2024
PHY 101: *Introductory Physics I*, Princeton University. Lecture on "Vibrational Motion and Mechanical Waves."
ART 3505: *Color*, Tyler School of Art and Architecture. Lecture on "The Physics of Light and Color."

Teaching Transcript Program 2022–2024
McGraw Center for Teaching and Learning, Princeton University
Five pedagogy workshops, guest lecturing with classroom observation, and the development of an original syllabus. Completed Fall 2024.

Reading Course Instructor 2019
University of Chicago, Chicago, IL
Designed and taught a reading course, *"Quantum Information and Quantum Chaos,"* for undergraduate physics majors.

Splash! Chicago Teacher 2018
University of Chicago, Chicago, IL
Designed and taught short courses introducing quantum mechanics and quantum computing to local high school students.

Graduate Teaching Assistant, University of Chicago 2017–2021
Introductory and honors sequences (**PHYS 121–123**, **PHYS 133**, **PHYS 142**), including their laboratory sections, *Physics for Future Presidents* (**PHSC 116–117**), and graduate quantum mechanics and string theory (**PHYS 341**, **PHYS 484**).

Undergraduate Teaching Assistant and Tutor, Colgate University 2015–2017
Introductory mechanics and physics (**PHYS 111**, **PHYS 232**), astronomy (**ASTR 102**), and *Energy and Sustainability* (**CORE 101**).

Publications

Undergraduate researchers marked with †.

1. B. Ouyang†, **J. Kudler-Flam**, "Logarithmic negativity typically equals exact entanglement cost" arXiv:2607.01320. Under Review.
2. **J. Kudler-Flam**, E. Witten, "Wormholes and Averaging over N" arXiv:2605.15180.
3. M. S. Klinger, **J. Kudler-Flam**, G. Satishchandran, "Generalized Entropy is von Neumann Entropy II: The

complete symmetry group and edge modes" arXiv:2601.07910. Accepted to Physical Review D.

4. Akash Vijay, Ayush Raj, **J. Kudler-Flam**, Benoît Vermersch, Andreas Elben, Laimei Nie, "Analytically Continuing the Randomized Measurement Toolbox" arXiv:2511.02912. Accepted to Physical Review Letters.
5. **J. Kudler-Flam**, E. Witten, "Emergent Mixed States for Baby Universes and Black Holes" arXiv:2510.06376. JHEP 05 090 (2026).
6. A. Herderschee, **J. Kudler-Flam**, "Stringy algebras, stretched horizons, and quantum-connected wormholes" arXiv:2510.01556. JHEP 04 117 (2026).
7. A. Blommaert, **J. Kudler-Flam**, E. Y. Urbach, "Absolute entropy and the observer's no-boundary state" arXiv:2505.14771. JHEP 11 113 (2025).
8. **J. Kudler-Flam**, V. Narovlansky, N. Sopenko, "Optimal bound on long-range distillable entanglement" arXiv:2504.02926. Phys. Rev. Lett. 135, 060403.
9. **J. Kudler-Flam**, G. Penington, "It costs nothing to teleport information into a black hole" arXiv:2504.01058. Int. J. Mod. Phys. D 34, 2543002. *Awarded second prize, 2025 Gravity Research Foundation Essay Competition.*
10. **J. Kudler-Flam**, K. Prabhu, G. Satishchandran, "Vacua and infrared radiation in de Sitter quantum field theory" arXiv:2503.19957. Under Review.
11. D.L. Danielson, **J. Kudler-Flam**, G. Satishchandran, R.M. Wald, "How to Minimize the Decoherence Caused by Black Holes" arXiv:2501.04773. Phys. Rev. D 112, 025012.
12. H.J. Chen[†] and **J. Kudler-Flam**, "Free Independence and the Noncrossing Partition Lattice in Dual-Unitary Quantum Circuits" arXiv:2409.17226. Phys. Rev. B 111, 014311.
13. X. Dong, **J. Kudler-Flam**, and P. Rath, "Entanglement Negativity and Replica Symmetry Breaking in General Holographic States" arXiv:2409.13009. JHEP 01 022 (2025).
14. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, "Algebraic Observational Cosmology" arXiv:2406.01669. Phys. Rev. D 114, 045007.
15. **J. Kudler-Flam**, S. Leutheusser, A. A. Rahman, G. Satishchandran, and A. Speranza, "A covariant regulator for entanglement entropy: proofs of the Bekenstein bound and QNEC" arXiv:2312.07646. Phys. Rev. D 111, 105001.
16. X. Dong, **J. Kudler-Flam**, and P. Rath, "A Modified Cosmic Brane Proposal for Holographic Renyi Entropy" arXiv:2312.04625. JHEP 06 120 (2024).
17. **J. Kudler-Flam**, S. Leutheusser, and G. Satishchandran, "Generalized Black Hole Entropy is von Neumann Entropy" arXiv:2309.15897. Phys. Rev. D 111, 025013.
18. **J. Kudler-Flam**, M. Nozaki, T. Numasawa, S. Ryu, and M.T. Tan, "Bridging two quantum quench problems: local joining quantum quench and Möbius quench, and their holographic dual descriptions," arXiv:2309.04665. JHEP 08 213 (2024).
19. **J. Kudler-Flam**, L. Nie, and A. Vijay, "Rényi mutual information in quantum field theory, tensor networks, and gravity" arXiv:2308.08600. JHEP 06 195 (2024).
20. G. Cipolloni and **J. Kudler-Flam**, "Non-Hermitian Hamiltonians Violate the Eigenstate Thermalization Hypothesis," arXiv:2303.03448. Phys. Rev. B 109, L020201. *Editor's Suggestion.*
21. Y. Liu, **J. Kudler-Flam**, and K. Kawabata, "Symmetry Classification of Typical Quantum Entanglement," arXiv:2301.07778. Phys. Rev. B 108, 085109.
22. Y. Liu, Y. Kusuki, R. Sohal, **J. Kudler-Flam**, and S. Ryu, "Multipartite entanglement in two-dimensional chiral topological liquids," arXiv:2301.07130. Phys. Rev. B 109, 085108.
23. **J. Kudler-Flam** and Y. Kusuki, "On Quantum Information Before the Page Time," arXiv:2212.06839. JHEP 05 078 (2023).
24. **J. Kudler-Flam**, "Rényi Mutual Information in Quantum Field Theory," arXiv:2211.01392. Phys. Rev. Lett. 130, 021603.
25. G. Cipolloni and **J. Kudler-Flam**, "Entanglement Entropy of Non-Hermitian Eigenstates and the Ginibre Ensemble," arXiv:2206.12438. Phys. Rev. Lett. 130, 010401.

26. **J. Kudler-Flam** and P. Rath, "Large and Small Corrections to the JLMS Formula from Replica Wormholes," arXiv:2203.11954. JHEP 08 189 (2022).
27. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, "Mixed-state entanglement and information recovery in thermalized states and evaporating black holes," arXiv:2112.00020. JHEP 01 064 (2023).
28. Y. Liu, R. Sohal, **J. Kudler-Flam**, and S. Ryu, "Multipartitioning topological phases by vertex states and quantum entanglement," arXiv:2110.11980. Phys. Rev. B 105, 115107.
29. S. Vardhan, **J. Kudler-Flam**, H. Shapourian, and H. Liu, "Bound entanglement in thermalized states and black hole radiation," arXiv:2110.02959. Phys. Rev. Lett. 129, 061602.
30. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, "Negativity Spectra in Random Tensor Networks and Holography," arXiv:2109.02649. JHEP 02 076 (2022).
31. **J. Kudler-Flam**, V. Narovlansky, and S. Ryu, "Distinguishing Random and Black Hole Microstates," arXiv:2108.00011. PRX Quantum 2, 040340.
32. **J. Kudler-Flam**, R. Sohal, and L. Nie, "Information Scrambling with Conservation Laws," arXiv:2107.04043. SciPost Phys. 12, 117 (2022).
33. **J. Kudler-Flam**, "Relative Entropy of Random States and Black Holes," arXiv:2102.05053. Phys. Rev. Lett. 126, 171603.
34. H. Shapourian, S. Liu, **J. Kudler-Flam**, and A. Vishwanath, "Entanglement negativity spectrum of random mixed states: A diagrammatic approach," arXiv:2011.01277. PRX Quantum 2, 030347.
35. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, "The quasi-particle picture and its breakdown after local quenches: mutual information, negativity, and reflected entropy," arXiv:2008.11266. JHEP 03 146 (2021).
36. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, "Entanglement of Local Operators and the Butterfly Effect," arXiv:2005.14243. Phys. Rev. Res. 3, 033182.
37. M. Asrat and **J. Kudler-Flam**, "$T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property," arXiv:2005.08972. Phys. Rev. D 102, 045009.
38. I. MacCormack, M. Tian Tan, **J. Kudler-Flam**, and S. Ryu, "Operator and entanglement growth in non-thermalizing systems: many-body localization and the random singlet phase," arXiv:2001.08222. Phys. Rev. B 104, 214202.
39. **J. Kudler-Flam**, Y. Kusuki, and S. Ryu, "Correlation measures and the entanglement wedge cross-section after quantum quenches in two-dimensional conformal field theory," arXiv:2001.05501. JHEP 04 074 (2020).
40. **J. Kudler-Flam**, L. Nie, and S. Ryu, "Conformal field theory and the web of quantum chaos diagnostics," arXiv:1910.14575. JHEP 01 175 (2020).
41. **J. Kudler-Flam**, H. Shapourian, and S. Ryu, "The negativity contour: a quasi-local measure of entanglement for mixed states," arXiv:1908.07540. SciPost Phys. 8, 063 (2020).
42. Y. Kusuki, **J. Kudler-Flam**, and S. Ryu, "Derivation of holographic negativity in AdS_3/CFT_2," arXiv:1907.07824. Phys. Rev. Lett. 123, 131603.
43. **J. Kudler-Flam**, M. Nozaki, S. Ryu, and M. Tian Tan, "Quantum vs. classical information: operator negativity as a probe of scrambling," arXiv:1906.07639. JHEP 01 031 (2020).
44. **J. Kudler-Flam**, I. MacCormack, and S. Ryu, "Holographic entanglement contour, bit threads, and the entanglement tsunami," arXiv:1902.04654. J. Phys. A: Math. Theor. 52 325401 (2019).
45. **J. Kudler-Flam** and S. Ryu, "Entanglement negativity and minimal entanglement wedge cross-sections in holographic theories," arXiv:1808.00446. Phys. Rev. D 99, 106014.
46. J. Chen and **J. Kudler-Flam**[†], "Laplacian growth & sandpiles on the Sierpinski gasket: limit shape universality and exact solutions," arXiv:1807.08748. Annales de l'Institut Henri Poincaré D 7 (4).

Selected Honors & Awards

Second Prize, Gravity Research Foundation Essay Competition 2025
Phi Beta Kappa Daniel H. Saracino Prize for Scholarship of Exceptional Merit, Colgate University 2017

Professional Activities

Journal Referee
Physical Review Letters, Physical Review E, Nature Physics, Journal of High Energy Physics, SciPost Physics, European Physical Journal Plus, European Physical Journal C, Journal of Mathematical Physics, Quantum

Organizer
Amherst Center for Fundamental Interactions (UMass Amherst) Workshop: "Black Holes and Horizons in Quantum Gravity" (Oct. 2026)
Institute for Advanced Study/Princeton Center for Theoretical Science Workshop: "Workshop on Quantum Aspects of Black Holes and Spacetime" (Dec 2025, page)
Princeton Center for Theoretical Science Workshop: "Random Physics" (May 2024, page)
Princeton Center for Theoretical Science Lunch Seminars (Fall 2023-Spring 2024)
Princeton Center for Theoretical Science Workshop: "Quantum Information, Dynamics and Ergodicity: From Many-Body Systems to Gravity" (Feb.-Mar. 2023, page)
Kadanoff Center for Theoretical Physics Journal Club (University of Chicago, page)
Kadanoff Center Group Meeting → Theory Group Meeting (University of Chicago)

Community

Prison Teaching Initiative Instructor (Princeton University, page) 2023 - Present
Big Brothers Big Sisters Mentor (Philadelphia, PA, page) 2022 - Present

Invited Talks

Wormholes and Averaging over N
- Kavli Institute for Theoretical Physics, Santa Barbara, CA (High Energy Theory / Gravitation Seminar, Jun. 2026, video)
- Simons Center for Geometry and Physics, Stony Brook, NY (Workshop on DSSYK Model: From Gravity to Many-Body Quantum Chaos, May 2026, video)

Progress report on large-N algebras and wormholes
- Bernoulli Center for Fundamental Studies, EPFL, Lausanne, Switzerland (Observers, wormholes and complex saddles in cosmology, May 2026)

Large fluctuations in the large-N limit
- Stanford University, Stanford, CA (SITP Monday Colloquium, Mar. 2026)
- University of Michigan, Ann Arbor, MI (LITP Workshop on Quantum Black Holes, Mar. 2026, video)

Generalized Entropy is von Neumann Entropy
- OIST, Okinawa, Japan (Qubits and Spacetime Unit Group Seminar, Mar. 2026, video)

Emergent Mixed States for Baby Universes and Black Holes
- Institute for Advanced Study, Princeton, NJ (Workshop on Quantum Aspects of Black Holes and Spacetime, Dec. 2025)
- Steklov Mathematical Institute, Moscow, Russia (Frontiers in Holography Workshop, Nov. 2025)
- Brandeis University, Waltham, MA (Quantum/Gravity Seminar, Oct. 2025)

University of Massachusetts, Amherst, MA (Amherst Center for Fundamental Interactions Seminar, Oct. 2025)
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Oct. 2025, video)

Stringy algebras, stretched horizons, and quantum-connected wormholes
Massachusetts Institute of Technology, Cambridge, MA (String/Gravity Theory Seminar, Nov. 2025)

Discussion of the Lorentzian Gravitational Path Integral
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Oct. 2025)

An Optimal Bound on Long-Range Distillable Entanglement
Castello dei Principi Capano, Pollica, Italy (Workshop on Universality in non-equilibrium quantum matter, Sep. 2025)
Texas Tech, Lubbock, TX (Quantum Information and Computing Seminar, Apr. 2025)

Absolute Entropy and the Observer's No-Boundary State
Perimeter Institute, Waterloo, Canada (QIQG 2025, Jun. 2025, video)
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Jun. 2025)

Entropy and the Hartle-Hawking State
Brown University, Providence, RI (Brown Theoretical Physics Center IDEA Seminar Series, Apr. 2025)

Comments on the identity matrix
Princeton University, Princeton, NJ (PCTS Year End Celebration, Mar. 2025)

Aspects of the Hartle-Hawking State
Cornell University, Ithaca, NY (LEPP Theory Seminar, Feb. 2025)

An Energy Efficient way to send Qubits into Black Holes
Institute for Advanced Study, Princeton, NJ (Physics Group Meeting, Feb. 2025)

Black Hole Entropy is von Neumann Entropy
University of Massachusetts, Amherst, MA (NE Gravity Workshop, Jan. 2025)
Trinity College, Dublin, Ireland (HMI Workshop: CFT and Holography for Heavy and Thermal States and Black Holes, June. 2024)

Vacua and Infrared Tails in de Sitter
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Nov. 2024)

Algebraic Observational Cosmology
University of Toronto, Toronto, Canada (Theoretical High-Energy Physics Seminar, May. 2024)
University of Chicago, Chicago, IL (Quantum Aspects of Cosmology Workshop, Mar. 2024)
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Mar. 2024)

Algebras and entropies for quantum field theory, black holes, and inflationary cosmology
Caltech University, Pasadena, CA (High Energy Theory Seminar, Mar. 2024)

Entropy differences in quantum field theory and a rigorous proof of the Bekenstein bound
Institute for Advanced Study, Princeton, NJ (Physics Group Meeting, Oct. 2023)

Horizons as eavesdroppers: decoherence from soft radiation
KTH Royal Institute of Technology, Stockholm, Sweden (Complex Systems and Biological Physics Seminars, May 2023)
Purdue University, West Lafayette, IN (Department of Physics & Astronomy Seminar, Apr. 2023, video)
Oxford University, Oxford, UK (Sondhi Group Meeting, Apr. 2023)
University of California Berkeley, Berkeley, CA (Berkeley Center for Theoretical Physics String Seminar, Apr. 2023,

video)
Stanford University, Stanford, CA (Stanford Institute for Theoretical Physics Seminar, Apr. 2023)
City College of the City University of New York, New York, NY (Theoretical Physics Seminar, Mar. 2023)

Explorations of Dissipative Quantum Chaos from Non-Hermitian Random Matrix Theory
Institute for Basic Science, Daejeon, Republic of Korea (Center for Theoretical Physics of Complex Systems Seminar, May 2023, video)

The Information-Disturbance Tradeoff and Horizons as Eavesdroppers
Institute for Advanced Study, Princeton, NJ (Quantum Aspects of Black Holes Meeting, Mar. 2023)

Rényi mutual information in quantum field theory and gravity
University of Pennsylvania, Philadelphia, PA (High Energy Theory Seminar, Feb. 2023)
Southeast University, Nanjing, China (2022 Forum for Young Scholars in Mathematics and Mathematical Physics, Dec. 2022)

Quantum Information before the Page Time
Institute for Advanced Study, Princeton, NJ (High Energy Theory Seminar, Sep. 2022, video)

Distinguishability and its use: from spin chains to black holes
Princeton University, Princeton, NJ (Princeton Center for Theoretical Science Annual Retreat, Sep. 2022)

On the Information Content of Black Hole Radiation
University of Amsterdam, Amsterdam, Netherlands (Institute of Physics String Theory Seminar, Mar. 2022)

Distinguishability in Random States, Eigenstates, and Gravity
Steklov Mathematical Institute, Moscow, Russia (Frontiers in Holography Workshop, Dec. 2021, video)
Institute for Research in Fundamental Sciences, Tehran, Iran (IPM Holography Virtual Seminar, Nov. 2021)
University of Illinois, Urbana Champaign, IL (Mathematical and Theoretical Physics Seminar, Oct. 2021)

Distinguishability of random states and the subsystem ETH
Harvard University, Cambridge, MA (Vishwanath Group Meeting, Jul. 2021)

Distinguishing Random States and Black Holes
Strings 2021 Gong Show, São Paulo, Brazil (Jun. 2021, video)

Entanglement Negativity Spectrum of Random Mixed States
Yukawa Institute for Theoretical Physics, Kyoto, Japan (Entanglement Meeting, Feb. 2021)

Universality of Entanglement Dynamics in Integrable and Chaotic Quantum Systems
University of Illinois Urbana Champaign, Champaign IL (Condensed Matter-AMO Journal Club, Oct. 2020)
Massachusetts Institute of Technology, Cambridge, MA (Condensed Matter Theory Seminar, Apr. 2020)

$T\bar{T}$, the entanglement wedge cross section, and the breakdown of the split property
Massachusetts Institute of Technology, Cambridge, MA (CTP Group Meeting, Aug. 2020)
Stanford University, Stanford, CA (SITP $T\bar{T}$ Group Meeting Seminar, Jun. 2020, video)

Entanglement, Gravity, and the Holographic Principle
Colgate University, Hamilton, NY (Physics & Astronomy Seminar, Jun. 2020)

Derivation of holographic negativity in AdS_3/CFT_2
Albert Einstein Institute, Potsdam, Germany (Gravity, Quantum Fields, & Information Seminar, Dec. 2019, video)

Strong shape theorems in cellular automata models on fractal graphs
AMS Special Sessions on Analysis and Geometry of Fractals, Riverside, CA (Nov. 2017)